Tanmay Tyagi

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Education

Netaji Subhas University of Technology — Dwarka, New Delhi
Bachelor of Technology — *Expected Aug. 2027*

Bharti Public School — New Delhi
Senior Secondary Education — *2021-2023*

Projects

OpenGL Rendering Engine | *C++, OpenGL, GLAD, Assimp* — June 2025 – July 2025

- Built modern OpenGL rendering pipeline using GLAD, Assimp, and stb_image.
- Applied Blinn–Phong lighting, fog effects, depth buffering, and camera system.
- Migrated the build system from Makefiles to CMake for cross-platform maintainability and cleaner dependency management.
- Debugged memory issues using Valgrind and AddressSanitizer.

Procedural Terrain Generation Engine | *C++, OpenGL, GLAD, OOP* — July 2025 – Oct 2025

- Developed a procedural terrain generation engine with chunk-based mesh generation for scalable real-time rendering.
- Implemented Perlin Noise and Value Noise from scratch to generate smooth heightmaps and natural terrain variation.
- Added Midpoint Displacement and Fault Formation algorithms to support multiple terrain generation techniques.
- Integrated the terrain system with an OpenGL renderer, free-fly camera, and custom GLSL vertex/fragment shaders.

Video Processing Pipeline | *C++, SDL3, FFmpeg, Multithreading, Scripting* — Feb 2025 – Mar 2025

- Implemented Triple Buffering and Producer-Consumer Pipeline for efficient video encoding-decoding.
- Handled cross-platform build dependencies (Linux/Windows) using Shell and Bash scripting.
- Achieved real-time Processing of videos with 4K 30fps resolution.
- Applied multiple post-processing effects such as Live Video to ASCII conversion by rendering glyphs using Streaming Textures.

GPU Accelerated Optimization Algorithm | *CUDA, OpenCL, NVIDIA Nsight* — Jan 2026 – Present

- Implemented Whale Optimization Algorithm on CPU and GPU.
- Wrote CUDA kernels using NVIDIA cuRAND for population initialization.
- Achieved approximately 140× speedup over CPU implementation through CUDA parallelization under local testing conditions.
- Designed complex benchmark fitness functions for AI/ML experiments.

Achievements

- Finalist, Meta PyTorch OpenEnv Hackathon — Scaler School of Technology, April, 2026 Qualified for the final round out of 52,000 participants.

Technical Skills

Languages: C/C++, CUDA, OpenCL, Python, Rust, JavaScript, TypeScript
Libraries & Frameworks: React.js, Express.js, React Native, std::thread, AVX2, OpenMP, stb_image
Graphics: OpenGL, WebGL, GLAD, GLFW, Assimp
Developer Tools: CMake, GDB, Valgrind, AddressSanitizer, gprof, Autocannon, VirtManager, NVIDIA Nsight Compute, Linux
Platforms: GitHub, Docker, DataGrip, Android Studio, Bruno, Postman, Visual Studio, Vim
Scripting Languages: Shell and Bash Scripting